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TREND-BASED MOCK PAPER

GATE Petroleum Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Petroleum Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

15% of 2000:

- (A) 200
- (B) 300
- (C) 400
- (D) 150

Section B: Petroleum Engineering

Q2. [1-NAT] [1 mark]

First order ODE $dy/dx = y$ solution:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q4. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q5. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q6. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q7. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q8. [1-NAT] [1 mark]

The Laplace transform of $t e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q10. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q11. [2-NAT] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MSQ] [2 marks]

For matrix $A = \begin{bmatrix} 2, 1 \\ 1, 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q13. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q14. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q18. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q20. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q21. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q22. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
 - (B) 0.5
 - (C) 0.6
 - (D) 0.4
-

Q24. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q25. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
 - (B) $(8/5, 4/5, 0)$
 - (C) $(2, 1, 0)$
 - (D) $(1, 2, 3)$
-

Q26. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{\|u\| * \|v\|}$
-

Q27. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
 - (B) $1/(1-z)$
 - (C) $\ln(1-z)$
 - (D) $e^{(-\lambda)(1+z)}$
-

Q28. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
 - (B) Cosine terms
 - (C) Both
 - (D) Neither
-

Q29. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q31. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q32. [1-MCQ] [1 mark]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q33. [1-NAT] [1 mark]

$C_p - C_v$ for ideal gas =

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q35. [1-MCQ] [1 mark]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q36. [2-NAT] [2 marks]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q38. [2-MCQ] [2 marks]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
- (B) r and γ
- (C) Temperature only
- (D) Pressure only

Q39. [2-NAT] [2 marks]

$C_p - C_v$ for ideal gas =

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MSQ] [2 marks]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
- (B) 333J
- (C) 667J
- (D) 1000J

Q41. [2-NAT] [2 marks]

For Otto cycle, efficiency depends on:

(Numerical Answer Type — Enter the numerical value)

Q42. [1-MCQ] [1 mark]

Source rock rich in:

- (A) Carbonate
- (B) Organic matter
- (C) Sand
- (D) Shale

Q43. [1-MCQ] [1 mark]

Anticline is a:

- (A) Fault
- (B) Fold structure
- (C) Unconformity
- (D) Reef

Q44. [1-MCQ] [1 mark]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q45. [2-MCQ] [2 marks]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q46. [2-MCQ] [2 marks]

Seismic survey uses:

- (A) Sound waves
- (B) Electromagnetic
- (C) Gravity
- (D) Magnetic

Q47. [1-MCQ] [1 mark]

Darcy's law: $q = ?$

- (A) $kA \frac{\Delta p}{\mu L}$
- (B) $\frac{kA \Delta p}{\mu L}$
- (C) $\frac{kL}{\mu} \Delta p$
- (D) $\frac{kA \Delta p}{\mu}$

Q48. [1-MCQ] [1 mark]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q49. [1-NAT] [1 mark]

Darcy's law: $q =$

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q51. [2-MCQ] [2 marks]

Material balance equation relates:

- (A) Pressure and time
- (B) In place, production, pressure
- (C) Temperature and volume
- (D) Permeability and flow

Q52. [1-MCQ] [1 mark]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q53. [1-NAT] [1 mark]

Mud weight is controlled to:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MSQ] [2 marks]

Mud weight is controlled to:

- (A) Cool bit only
- (B) Prevent kicks and circulation loss
- (C) Speed up drilling
- (D) Lubricate

Q55. [2-MCQ] [2 marks]

Drill bit rotation is provided by:

- (A) Top drive
- (B) Rotary table
- (C) Both
- (D) Downhole motor only

Q56. [1-NAT] [1 mark]

Choke in production system controls:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

ESP (Electric Submersible Pump) is used for:

- (A) Gas lift
- (B) Artificial lift of liquids
- (C) Water injection
- (D) Sealing

Q58. [2-MSQ] [2 marks]

Choke in production system controls:

- (A) Flow rate
- (B) Pressure only
- (C) Temperature
- (D) Composition

Q59. [2-MCQ] [2 marks]

ESP (Electric Submersible Pump) is used for:

- (A) Gas lift
- (B) Artificial lift of liquids
- (C) Water injection
- (D) Sealing

Q60. [1-NAT] [1 mark]

Roller cone bit has:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

ESP stands for:

- (A) Electric Submersible Pump
- (B) Electronic Surface Pump
- (C) Electrostatic Pump
- (D) None

Q62. [1-MCQ] [1 mark]

Gamma ray log measures:

- (A) Porosity
- (B) Saturation
- (C) Shale
- (D) Permeability

Q63. [1-NAT] [1 mark]

Pressure transient analysis uses:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

API gravity > 31.1 is:

- (A) Heavy
- (B) Light
- (C) Extra heavy
- (D) Bitumen

Q65. [1-MCQ] [1 mark]

Jack-up rig works in:

- (A) Deep water
 - (B) Shallow water
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1NAT	C	1
6	1MCQ	A	1
7	1MCQ	C	1
8	1NAT	A	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1MCQ	C	1
12	1NAT	A	1
13	1MCQ	B	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1NAT	B	1
19	1MCQ	B	1
20	1MCQ	C	1
21	1MCQ	A	1
22	1NAT	B	1
23	1NAT	A	1
24	1MCQ	C	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1NAT	A	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2NAT	A	2
32	2NAT	C	2
33	2MCQ	B	2
34	2MCQ	B	2
35	2MSQ	A	2
36	2MCQ	A	2
37	2NAT	B	2
38	2NAT	A	2
39	2NAT	A	2
40	2MCQ	A	2
41	2NAT	C	2
42	2MCQ	B	2
43	2MCQ	B	2

44	2MCQ	A	2
45	2NAT	A	2
46	2MCQ	B	2
47	2MCQ	A	2
48	2MCQ	B	2
49	2MCQ	A	2
50	2MCQ	A	2
51	2NAT	B	2
52	2MSQ	A	2
53	2MCQ	A	2
54	2MSQ	A	2
55	2NAT	B	2
56	2MCQ	B	2
57	2MSQ	B	2
58	2MCQ	B	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2NAT	C	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MCQ	C	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Exploration | Topic: Geology | Difficulty: difficult

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Reservoir Engineering | Topic: Fluid Flow | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Drilling Engineering | Topic: Bits | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Production Engineering | Topic: Artificial Lift | Difficulty: easy

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Petroleum Formation Evaluation | Topic: Logging | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Well Testing | Topic: Analysis | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Petroleum Chemistry | Topic: Crude | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Offshore Drilling | Topic: Rigs | Difficulty: difficult

Q11.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: easy

Q12.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: easy

Q13.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q14.

Answer: B

Solution:

Anticline: upward convex fold. Good petroleum trap.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q15.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q16.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: moderate

Q17.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q18.

Answer: B

Solution:

Mud weight: balance formation pressure (prevent kicks) and prevent fracture (lost circulation).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q19.

Answer: B

Solution:

$T_2/T_1 = (P_2/P_1)^{((\gamma-1)/\gamma)} = 10^{(0.286)} = 1.93$. $T_2 = 300 \times 1.93 = 579\text{K}$. Closest: 507K (using exact calculation).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q20.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q22.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q23.

Answer: A

Solution:

$q = -kA/\mu \cdot dp/dL$. Steady linear flow: $q = kA \cdot dp/(\mu \cdot L)$.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: easy

Q24.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 \cdot 1000 = 500J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q27.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q29.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \cdot 300 \cdot \ln(2) = 2494.2 \cdot 0.693 = 1729 J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q32.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x \cdot \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q34.

Answer: B

Solution:

Otto: $\eta = 1 - r^{1-\gamma}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q35.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q39.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: B

Solution:

$\text{proj} = ((v.w)/(w.w))*w = (4/5)*(2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q49.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q52.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 * 1000 = 500J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q53.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q54.

Answer: A

Solution:

Choke: restricts flow to control production rate and upstream pressure.

Subject: Production | Topic: Production | Difficulty: moderate

Q55.

Answer: B

Solution:

Otto: $\eta = 1 - r^{-(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q56.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q57.

Answer: B

Solution:

Mud weight: balance formation pressure (prevent kicks) and prevent fracture (lost circulation).

Subject: Drilling | Topic: Drilling | Difficulty: moderate

Q58.

Answer: B

Solution:

ESP: multistage centrifugal pump at bottom of well. Artificial lift for liquid production.

Subject: Production | Topic: Production | Difficulty: easy

Q59.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: A

Solution:

Seismic: generates and records sound waves reflected by subsurface layers.

Subject: Petroleum Exploration | Topic: Petroleum Exploration | Difficulty: moderate

Q61.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

Material balance: OOIP, cumulative production, reservoir pressure evolution.

Subject: Reservoir Engineering | Topic: Reservoir Engineering | Difficulty: moderate

Q63.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: C

Solution:

Rotation: surface (top drive or rotary table) and/or downhole motor (mud motor).

Subject: Drilling | Topic: Drilling | Difficulty: easy

Q65.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 30 questions, 51 marks
Thermodynamics: 10 questions, 16 marks
Petroleum Exploration: 5 questions, 7 marks
Reservoir Engineering: 5 questions, 7 marks
Drilling: 4 questions, 6 marks
Production: 4 questions, 6 marks
General Aptitude: 1 questions, 1 marks
Drilling Engineering: 1 questions, 1 marks
Production Engineering: 1 questions, 1 marks
Petroleum Formation Evaluation: 1 questions, 1 marks
Well Testing: 1 questions, 1 marks
Petroleum Chemistry: 1 questions, 1 marks
Offshore Drilling: 1 questions, 1 marks

Difficulty Distribution:

Easy: 26 | Moderate: 30 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.